

THERMODYNAMIC CHARACTERISTICS OF AMMONIUM ION SOLVATION IN N-METHYLPYRROLIDONE–WATER SOLUTIONS

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Based on the standard values of the heat capacity $\overline{C_{p,i}^\circ}$ of an ammonium ion, the values of the change in heat capacity during the solvation $\Delta_{\text{solv.}}C_{p,i}^\circ$ of an ammonium ion in a mixed solvent N-methylpyrrolidone (MP)-water at 298.15 K are determined. The magnitude of the ammonium ion is represented as the sum of four contributions: electrostatic $\Delta_{\text{el.}}C_{p,i}^\circ$, cavity formation $\Delta_{\text{cav.}}C_{p,i}^\circ$, structural changes in the solvent $\Delta_{\text{str.}}C_{p,i}^\circ$, and specific ion-solvent interactions $\Delta_{\text{sp.int.}}C_{p,i}^\circ$. The values of the contributions are calculated based on experimental data and model representations. The calculation results are presented in the table.

The change in heat capacity during solvation $\Delta_{\text{solv.}}C_{p,i}^\circ$ of an ammonium ion in a mixed solvent MP – water, MP and water at 298.15 K and contributions to this value caused by specific interactions, electrostatic interactions, cavity formation, and changes in the solvent structure

J/mol·K	X_{MP}						
	0,00	0,10	0,33	0,50	0,75	0,90	1,00
$\Delta_{\text{solv.}}C_{p,i}^\circ$	-62	-78	-80	-41	-17	7	12
$\Delta_{\text{sp.int.}}C_{p,i}^\circ$	30	4	2	-7	-5	-8	-9
$\Delta_{\text{el.}}C_{p,i}^\circ$	-218	-225	-236	-319	-310	-285	-295
$\Delta_{\text{cav.}}C_{p,i}^\circ$	41	82	97	92	109	116	134
$\Delta_{\text{str.}}C_{p,i}^\circ$	85	60	57	193	189	170	155

The dependences of the obtained values on the composition of the mixture are discussed in connection with the structure of the mixed solvent.

The research was carried out with the financial support of the Ministry of Science and Higher Education of the Russian Federation within the framework of the scientific project of the laboratory "Laboratory of Ion Materials" (LIM), project no. FSSM-2024-0006.